Numerical Verification of Theorem 3.1
(Bertsimas & Tsitsiklis)

Theorem 3.1.
Let $\mathbf{x}$ be a basic feasible solution with basis matrix B and reduced costs $\bar{c}_j = c_j - \mathbf{c}_B^\top B^{-1} A_j$. (a) If $\bar{\mathbf{c}} \geq 0$, then $\mathbf{x}$ is optimal. (b) If $\mathbf{x}$ is optimal and nondegenerate, then $\bar{\mathbf{c}} \geq 0$.

1 The problem

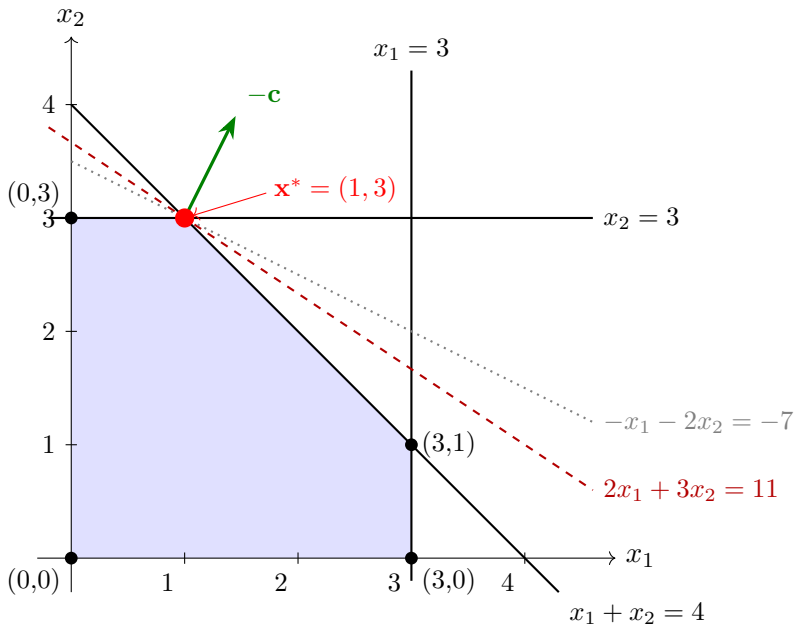
$$\begin{aligned} \min \quad & -x_1 - 2x_2 \\ \text{s.t.} \quad & x_1 + x_2 \leq 4, \quad x_1 \leq 3, \quad x_2 \leq 3, \quad x_1, x_2 \geq 0. \end{aligned}$$

Adding slacks $s_1, s_2, s_3 \geq 0$ gives the standard form $A\mathbf{x} = \mathbf{b}$, $\mathbf{x} \geq 0$, with variables $(x_1, x_2, s_1, s_2, s_3)$:

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 4 \\ 3 \\ 3 \end{pmatrix}, \qquad \mathbf{c} = (-1, -2, 0, 0, 0)^\top.$$

For each basis we compute $\mathbf{p}^\top = \mathbf{c}_B^\top B^{-1}$ (i.e. solve $B^\top \mathbf{p} = \mathbf{c}_B$) and then $\bar{c}_j = c_j - \mathbf{p}^\top A_j$.

2 The polyhedron



Objective values at the vertices: $(0,0)$: 0, $(3,0)$: -3, $(3,1)$: -5, $(0,3)$: -6, $(1,3)$: -7. So $\mathbf{x}^* = (1,3)$ is the unique optimum.

3 Part (a): $\bar{\mathbf{c}} \geq 0$ certifies optimality

At $\mathbf{x}^* = (1,3)$: $s_1 = 0$, $s_2 = 2$, $s_3 = 0$. Basis $\{x_1, x_2, s_2\}$:

$$B = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{x}_B = B^{-1}\mathbf{b} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}, \quad \mathbf{p} = \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix}.$$

Nonbasic reduced costs:

$$\bar{c}_{s_1} = 0 - \mathbf{p}^\top e_1 = 1, \qquad \bar{c}_{s_3} = 0 - \mathbf{p}^\top e_3 = 1.$$

Hence $\bar{\mathbf{c}} = (0,0,1,0,1) \geq 0$, and by (a) $\mathbf{x}^*$ is optimal, which agrees with the vertex table.

Contrast: a non-optimal BFS. At $(3,1)$, basis $\{x_1, x_2, s_3\}$, $\mathbf{x}_B = (3,1,2)$ (nondegenerate), $\mathbf{p} = (-2,1,0)$ and $\bar{\mathbf{c}} = (0,0,2,-1,0)$. Since $\bar{c}_{s_2} = -1 < 0$, increasing s_2 (moving along the edge $x_1 + x_2 = 4$ toward $(1,3)$) lowers the cost by 1 per unit. This is consistent with (b): this BFS is nondegenerate and has a negative reduced cost, so it cannot be optimal.

4 Part (b): nondegeneracy matters

The basis in Section 3 is nondegenerate ($\mathbf{x}_B = (1,3,2) > 0$) and optimal, and indeed $\bar{\mathbf{c}} \geq 0$, as (b) states.

To see why the hypothesis is needed, add the *redundant* constraint $2x_1 + 3x_2 \leq 11$ (slack s_4), which is tight only at $(1,3)$ (dashed line). The feasible set and the optimum are unchanged, but now $m = 4$ and four constraints are active at $(1,3)$, so every BFS at this point is **degenerate**. Take the basis $\{x_1, x_2, s_2, s_3\}$:

$$\mathbf{x}_B = (1,3,2,\mathbf{0}), \qquad \mathbf{p} = (1,0,0,-1), \qquad \bar{\mathbf{c}} = (0,0,\,\mathbf{-1},\,0,\,0,\,1).$$

Here $\bar{c}_{s_1} = -1 < 0$, yet $\mathbf{x}^*$ is optimal. The direction associated with s_1 is infeasible: with $s_3 = 0$ basic, it would push s_3 negative, so the step length is $\theta^* = 0$.

A different basis for the same point, $\{x_1, x_2, s_2, s_4\}$, gives $\bar{\mathbf{c}} = (0,0,1,0,1,0) \geq 0$. So the optimality certificate depends on the choice of basis: for a degenerate optimal $\mathbf{x}$, some basis has $\bar{\mathbf{c}} \geq 0$ (this is what the simplex method finds), but not necessarily every basis.

5 Summary

Point	Basis	Degenerate?	$\bar{\mathbf{c}}$	Optimal?
$(1,3)$	x_1, x_2, s_2	no	$(0,0,1,0,1) \geq 0$	yes
$(3,1)$	x_1, x_2, s_3	no	$(0,0,2,-1,0)$	no
$(1,3)$, extra row	x_1, x_2, s_2, s_3	yes	$(0,0,-1,0,0,1)$	yes
$(1,3)$, extra row	x_1, x_2, s_2, s_4	yes	$(0,0,1,0,1,0) \geq 0$	yes